

Notes: Behn, Haselmann, & Wachtel (JF, 2016)

Procyclical Capital Regulation and Lending

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1 Big picture

- **Question.** Does model-based, risk-sensitive bank capital regulation make lending more procyclical when credit risk rises?
- **Setting.** German banks around the September 2008 Lehman shock. Germany is useful because some banks had adopted Basel II internal ratings-based regulation (IRB), while others used the standard approach (SA), and IRB banks had only partially rolled out IRB across their loan portfolios.
- **Key institutional contrast.** Under IRB, capital charges depend on internal credit-risk estimates such as probability of default (PD). Under SA, most unrated corporate loans have fixed risk weights, so capital charges do

not mechanically rise when PDs rise.

- **Data.** German credit-register data on loans of at least EUR 1.5 million, including lender, borrower, loan amount, collateral, PD, risk-weighted assets (RWAs), and the regulatory approach used for the loan. The main pre-period is 2008Q1-2008Q2 and the post-period is 2008Q4-2011Q3.
- **Core shock.** The Lehman collapse is treated as an exogenous credit-risk shock. It sharply changed real-sector expectations and forced IRB banks to update internal PDs and risk weights.
- **Main result.** Capital requirements for IRB loans rose by about 0.5 percentage points. Banks reduced IRB loans by roughly 2.1 to 3.9 percentage points more than comparable SA loans to the same firm.
- **Firm-level effect.** Firms with a higher pre-shock share of IRB loans experienced larger reductions in total bank borrowing. A movement from the 25th to the 75th percentile of IRB exposure implies an additional reduction of roughly 3.0 to 4.6 percentage points in aggregate borrowing.
- **Mechanism.** Banks mostly adjusted quantities, not prices: loan balances fell, but interest rates and firm-level capital costs did not rise much.
- **Economic takeaway.** Risk-sensitive microprudential rules can strengthen individual bank safety, but when risk estimates rise in a downturn, the same rules can force many banks to deleverage at once and amplify the downturn.

2 Data and measurement

2.1 Basel II regulatory approaches

- **Standard approach (SA).** Similar to Basel I. For most German corporate loans without external ratings, the uncollateralized part receives a fixed 100% risk weight.
- **Internal ratings-based approach (IRB).** Capital charges are based on the bank's own model estimates: PD, loss given default, exposure at default, and maturity.
- **Why this matters.** If the macroeconomy deteriorates, estimated PDs rise. Under IRB, risk weights and capital charges rise. Under SA, capital charges are largely unchanged.

Interpretation. IRB creates a state-contingent capital requirement. SA creates a fixed requirement. The paper compares lending responses when the same aggregate shock affects only one regulatory approach mechanically.

2.2 German phase-in of IRB

- Banks that adopted IRB did not switch every loan at once. They submitted implementation plans in 2006 and gradually moved entire portfolios into IRB.
- The switch applied to loan portfolios, not hand-picked individual loans. Once a portfolio was moved to IRB, all loans in that portfolio were treated under IRB.
- At the Lehman shock, IRB banks still had both IRB and SA loan pools.

Interpretation. This creates within-bank variation: the same IRB bank can have both IRB loans and SA loans. That lets the paper separate regulation from bank-level supply shocks.

2.3 Loan and firm samples

- The credit-register sample contains 1,825 banks: 1,784 SA banks and 41 IRB banks.
- Loan-level observations with identified regulatory approach include 92,466 loans from SA banks, 29,083 SA loans from IRB banks, and 61,417 IRB loans from IRB banks.
- The firm matched sample links the credit register to Bureau van Dijk Amadeus balance-sheet data for 7,778 firms.
- Key loan variables include loan size, collateral ratio, relationship-lender indicator, relationship length, PD, risk weight, interest rate, and changes around the shock.
- Key firm variables include assets, ROA, leverage, sales, total loans, firm capital cost, and the change in total firm borrowing.

Interpretation. The data are unusually powerful because they contain not just loan amounts, but also the capital-regulatory treatment and the internal risk estimate behind the capital charge.

2.4 Event windows and outcomes

- **Pre-event period:** 2008Q1-2008Q2.
- **Shock period:** 2008Q3, when Lehman failed.
- **Post-event period:** 2008Q4-2011Q3.
- Main outcome at the loan level: $\Delta \log(loans)_{ij}$, the change in loan size from bank j to firm i .
- Main outcome at the firm level: $\Delta \log(firm\ loans)_i$, the change in the firm's total bank borrowing.

Interpretation. Collapsing quarterly data into pre- and post-event averages avoids serial-correlation problems and

turns the exercise into a clean before-after comparison around the shock.

3 Models and empirical specifications

3.1 Test 1: IRB banks versus SA banks

$$\Delta \log(\text{loans})_{ij} = \alpha_i + \beta \text{ShareIRB}_j + X'_{ij} \gamma + \varepsilon_{ij}.$$

- ShareIRB_j is the share of bank j 's loans that are subject to IRB.
- Firm fixed effects α_i compare different lenders to the same borrower, absorbing firm-specific loan-demand shocks.

Interpretation. A negative β means that the same firm experiences a larger lending cut from banks with more IRB exposure.

3.2 Test 2: SA loans from IRB banks versus SA loans from SA banks

- This test restricts the sample to SA loans.
- It asks whether IRB banks reduce their SA loans more than SA banks.

Interpretation. If the main result were only about large IRB banks being different, SA loans from IRB banks should also fall more. If the result is truly about model-based capital charges, the effect should be weaker or absent for SA loans.

3.3 Test 3: IRB loans versus SA loans within IRB banks

$$\Delta \log(\text{loans})_{ij} = \alpha_i + \alpha_j + \delta \text{IRBLoan}_{pj} + X'_{ij} \gamma + \varepsilon_{ij}.$$

- $\text{IRBLoan}_{pj} = 1$ if the loan pool p of bank j is under IRB.
- Firm fixed effects absorb demand shocks.
- Bank fixed effects absorb bank-level supply shocks.

Interpretation. This is the strongest loan-level test. It compares the same firm borrowing from IRB banks, where one loan is under IRB and another is still under SA.

3.4 Within-IRB SME discount test

$$\Delta \log(\text{loans})_{ij} = \alpha_i + \alpha_{pj} + \zeta [\text{IRBLoan}_{pj} \times \log(\text{FirmSales}_i)] + \varepsilon_{ij}.$$

- Basel II gives SMEs a capital-charge discount when turnover is below EUR 50 million.
- For the same increase in PD, risk weights rise less for smaller firms.
- Bank-by-loan-pool fixed effects α_{pj} control for differences between a bank's IRB and SA pools.

Interpretation. A negative interaction means larger firms, which do not benefit from the SME discount, suffer larger cuts in IRB lending.

3.5 Firm-level borrowing regression

$$\Delta \log(\text{firm loans})_i = \alpha + \beta \overline{\text{IRBLoan}}_i + X'_i \gamma + \varepsilon_i.$$

- $\overline{\text{IRBLoan}}_i$ is the share of firm i 's loans subject to IRB before the shock.
- The paper also uses $\overline{\text{IRBLoan}}_i^*$, the share of loans from IRB banks that are subject to IRB, in a restricted sample with both IRB and SA loans from IRB banks.

Interpretation. The loan-level results show the direct bank response. The firm-level results ask whether firms can replace lost IRB credit with other bank credit.

3.6 Demand-adjusted firm-level coefficient

$$\tilde{\beta} = \hat{\beta}^{OLS} - \left(\hat{\beta}_{loan}^{OLS} - \hat{\beta}_{loan}^{FE} \right) \frac{\sigma_L^2}{\sigma_F^2}.$$

Interpretation. This adjustment uses the gap between loan-level OLS and firm-fixed-effect estimates to approximate unobserved firm demand shocks in the firm-level regression.

4 Identification and empirical design

4.1 What identifies the main effect

- The shock is a sudden post-Lehman rise in credit risk, not a gradual bank choice.
- IRB loans are exposed because PD-based risk weights rise.
- SA loans are less exposed because risk weights are fixed.

- Firm fixed effects absorb borrower demand shocks.
- Bank fixed effects absorb lender supply shocks.
- The strongest comparison uses loans to the same borrower from IRB banks, where one loan is under IRB and another is under SA.

Interpretation. Identification comes from comparing loans that face the same macro shock and similar borrower demand, but different capital-charge sensitivity.

4.2 Why the phase-in matters

- If all IRB banks had moved every loan into IRB at once, IRB status would be almost the same as bank identity.
- The gradual rollout means IRB status differs across loan portfolios within the same bank.
- Table II shows that portfolio size predicts IRB classification, but loan size, collateral, relationship-lender status, and relationship length do not strongly predict classification.

Interpretation. Portfolio-level phase-in turns Basel II adoption into a useful quasi-experiment. Remaining selection concerns are addressed with portfolio controls and the SME-discount test.

4.3 What the paper cannot fully prove

- The study cannot randomize IRB assignment; it relies on the institutional rollout and controls.
- Firm-level demand is harder to absorb than loan-level demand, so the firm-level analysis uses restrictions and demand-adjusted estimates.
- The setting is Germany during the financial crisis, so external validity depends on how similar other banking systems and capital rules are.

Interpretation. The paper is strongest on loan supply within the German credit-register setting. The broader macroprudential implication is persuasive but not a full general-equilibrium estimate.

5 Tables and figures

5.1 Figures 1 and 2: the shock, lending, and model-based RWAs

Read.

- Figure 1 shows the real shock: GDP falls sharply after Lehman, lending standards tighten, and aggregate lending stops rising.
- Figure 2 shows the regulatory mechanism: when holding portfolios constant, the RWA-to-loan ratio rises after the shock, consistent with IRB risk weights increasing.

Economic message. The economy deteriorates first, then IRB capital charges rise mechanically with PDs. The paper's mechanism is not just banks becoming cautious; it is risk-sensitive regulation converting bad news into higher required capital.

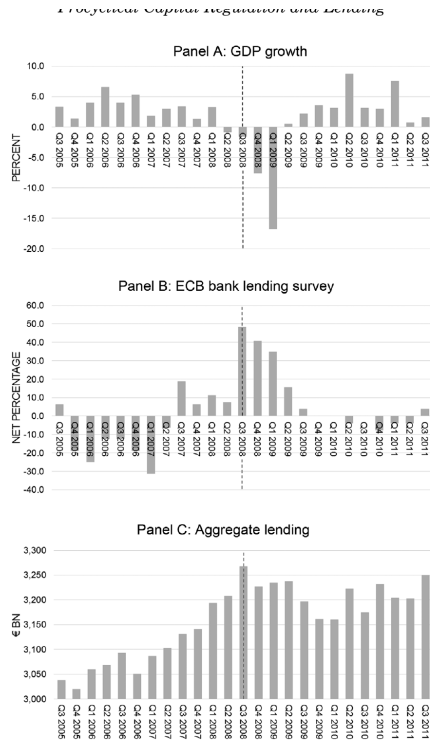


Figure 1. The credit risk shock and the German economy. Panel A shows the quarterly change in real GDP at an annual rate (calendar and seasonally adjusted). Panel B shows the responses of German banks to the following question from the ECB's bank lending survey: "Please indicate how you expect your bank's credit standards as applied to the approval of loans or credit lines to enterprises to change over the next three months." The values displayed are the proportion of respondents that expect a tightening minus the proportion that expect an easing of credit standards. Panel C shows the aggregate amount of loans (in billion euros) extended by all German banks to corporate borrowers and households. The dashed vertical line indicates the credit risk shock in September 2008. (Source: German Federal Statistical Office, European Central Bank, and Deutsche Bundesbank.)

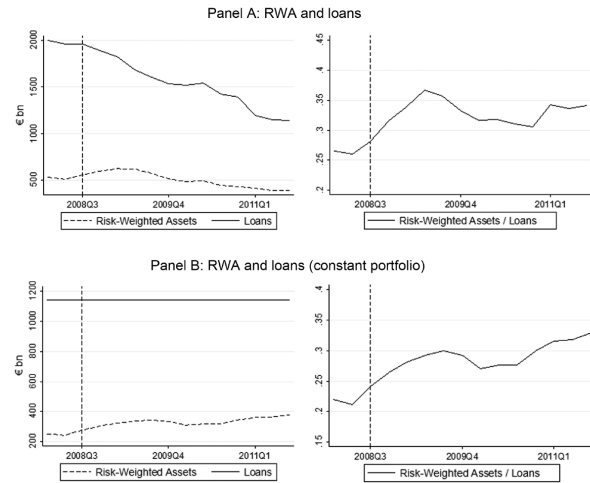


Figure 2. Aggregate risk-weighted assets (RWAs) and IRB loans. Panel A shows aggregate loans from the credit register and the corresponding RWAs. The series includes only those lending relationships that existed prior to the credit risk shock (i.e., we do not include lending relationships that were originated after the event). The left-hand graph shows loans and RWAs for these loans, and the right-hand graph depicts the ratio of RWAs to loans. Panel B shows RWAs and loans assuming a constant portfolio of loans. We include all loans that exist over the entire sample period and calculate the risk weight for each loan at each point in time. We calculate hypothetical RWAs by multiplying the loan amount of 2008Q3 with the risk weight for the respective period. The aggregate of the hypothetical RWAs and the (constant) loan total are shown in the left graph. The right graph shows the ratio between the two series.

5.2 Tables I and II: sample construction and IRB selection

Read.

- Table I shows that IRB banks are much larger than SA banks and have lower equity ratios. IRB loans have lower initial risk weights but experience larger increases in risk weights and larger declines in loan size.
- Table II shows the most important predictor of whether a loan is already in IRB is the bank's portfolio share in that sector; ordinary loan characteristics are not the main selection drivers.

Economic message. The tables support the view that banks first moved large portfolios to IRB. Because larger portfolios appear to be reduced less, this selection works against finding the paper's main negative IRB effect.

5.3 Table III: main loan-level results

Read.

- Test 1: banks with a higher IRB share cut lending more to the same firm. The implied IRB-bank effect is about 3.1 to 5.0 percentage points.
- Test 2: when the sample is restricted to SA loans, the effect mostly disappears once bank characteristics are controlled for.
- Test 3: within IRB banks, IRB loans fall 2.1 to 3.9 percentage points more than SA loans to the same firm.

Economic message. This is the paper's core evidence: the crunch is tied to the regulatory treatment of the loan, not only to the identity of the bank or borrower.

5.4 Figure 3 and Table IV: risk-weight formula and extensive margin

Read.

- Figure 3 shows that the risk-weight schedule rises with PD, and that SMEs receive a lower risk-weight mapping.
- Table IV asks whether IRB loans are more likely to disappear entirely. The IRB-loan exit coefficient is positive in the simplest specification, but weaker after controls and fixed effects.

Table I
Summary Statistics

Panel A shows summary statistics (mean and standard deviation, *S.D.*) for banks that introduced the internal ratings-based approach (IRB banks) and banks that retained the standard approach (SA banks). Panel B shows statistics for all loans for which we were able to identify the regulatory approach and includes information on risk-weighted assets (RWAs) and the probability of default (PD). Panel C shows statistics for the matched sample of firms for which we obtained balance sheet data and aggregate loans from the credit register. All level variables are measured prior to the credit risk shock, while changes indicate the difference between the average for the pre-event period (2008Q1 to 2008Q4) and the average for the postevent period (2008Q4 to 2011Q3).

Panel A: Bank-Level Variables					
	1,784 SA Banks		41 IRB Banks		
	Mean	S.D.	Mean	S.D.	
<i>Share IRB</i> —share of loans subject to IRB	0	0	0.620	0.371	
<i>Bank assets</i> —in millions of €	1,080	2,580	138,000	307,000	
<i>Bank equity ratio</i> —equity to assets	0.067	0.051	0.046	0.029	
<i>Bank ROA</i> —return on assets	0.006	0.012	0.006	0.010	

Panel B: Loan-Level Variables					
	SA Banks		IRB Banks		
	Mean	S.D.	Mean	S.D.	
<i>Loan size</i> —in millions of €	8,843	22,405	19,068	43,798	25,722 50,498
<i>Collateral ratio</i> —collateral to loan size	0.029	0.161	0.131	0.333	0.169 0.351
<i>Relationship leader</i> —indicator variable (see text)	0.470	0.499	0.545	0.498	0.468 0.499
<i>Relationship length</i> —in quarters	19.1	17.4	17.7	17.2	18.8 17.5
<i>PD</i> —probability of default	—	—	0.024	0.050	0.015 0.052
<i>Risk weight</i> —RWA to loan size	0.602	0.373	0.612	0.432	0.435 0.449
<i>Interest rate</i> —imputed (see text)	0.087	0.055	0.089	0.058	0.098 0.063
<i>Δ Log (loans)</i> —change in log of loan size	-0.012	0.359	-0.039	0.459	-0.075 0.569
<i>Δ PD (incl)</i> —change in PD, excl. defaults	—	—	0.005	0.047	0.006 0.039
<i>Δ PD (excl)</i> —change in PD, incl. defaults	—	—	0.021	0.100	0.030 0.126
<i>Δ Risk weight</i> —change in Risk weight	-0.014	0.180	-0.007	0.284	0.067 0.461
<i>Δ Interest rate</i> —change in interest rate	-0.017	0.036	-0.015	0.032	-0.017 0.037
Portfolio variables					
<i>Portfolio share</i> —see text	0.002	0.007	0.059	0.057	0.096 0.101
<i>Portfolio PD</i> —see text	—	—	0.022	0.018	0.017 0.016

(Continued)

Table I—Continued

Panel C: Firm-Level Variables		
	Firms (7,778 Obs.)	
	Mean	S.D.
<i>Firm assets</i> —in millions of €	133.4	347.9
<i>Firm ROA</i> —return on assets	0.065	0.093
<i>Firm leverage</i> —debt to assets	0.135	0.141
<i>Firm sales</i> —in millions of €	116.9	395.4
<i>Firm loans</i> —in millions of €	22.7	67.5
<i>Firm capital cost</i> —interest paid to firm loans	0.0829	0.0712
<i>Δ Log(firm loans)</i> —change in log of firm loans	-0.078	0.309
<i>Δ Capital cost</i> —change in firm capital cost	-0.0011	0.0201

Table II
Determinants of Loan Classification

The dependent variable in all regressions is an indicator for the regulatory approach used for the loan in 2008Q1. It is equal to one if the loan from bank *j* to firm *i* is subject to IRB and equal to zero if the loan is subject to the standard approach. *Log(loan size)* is equal to the logarithm of average loan size for the bank-firm relationship over the sample period (2008Q1 to 2011Q3). The remaining variables are defined in Table I. The sample includes all loans from IRB banks in columns 1 to 3 and the loans to firms with multiple lending relationships used for identification in Test 3 in columns 4 to 6. Robust standard errors adjusted for clustering at the bank and firm level are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: <i>IRB loan</i>					
	All Firms			Multiple Lending Relationships		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Portfolio share</i>	0.509* (0.297)	0.402** (0.271)	0.765** (0.305)	0.774** (0.278)		
<i>Portfolio PD</i>	-0.546 (1.105)	-0.517 (1.126)	-0.638 (0.868)	-0.596 (0.975)		
<i>Log (loan size)</i>		0.015 (0.012)	0.014 (0.011)	0.016 (0.010)	0.015 (0.010)	
<i>Collateral ratio</i>		0.084 (0.094)	0.087 (0.095)	0.093 (0.077)	0.096 (0.077)	
<i>Relationship leader</i>		-0.086 (0.037)	-0.038 (0.037)	-0.018 (0.050)	-0.022 (0.050)	
<i>Relationship length</i>		0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)	
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	90,500	90,500	90,500	27,620	27,620	27,620
<i>R</i> ²	0.849	0.850	0.851	0.553	0.558	0.560

Table III
Lending and the Regulatory Approach

The table shows the relationship between the decline in loan size and the regulatory approach used by the bank. We take the credit risk shock in late 2008 as an event and collapse all quarterly data for a given loan into a single pre- and postevent period. Data are restricted to (a) loans that are larger than €1.5 million (b) loans from commercial, state, or cooperative banks that are subject to the Basel II capital regulation, and (c) loans that have an observation in both the pre- and postshock period. The dependent variable is the difference in *Log (loans)* between the pre- and postevent periods. In columns 1 to 3 we use observations for firms that have at least one loan from an SA bank and one loan from an IRB bank or loans from at least two distinct IRB banks (Test 1). In columns 4 to 6 the sample is restricted to SA loans and includes only firms that have at least one SA loan from an SA bank and at least one SA loan from an IRB bank or SA loans from at least two distinct IRB banks (Test 2). Finally, columns 7 to 9 include only loans from IRB banks and only firms that have at least one SA loan and at least one IRB loan from an IRB bank (Test 3). The IRB impact at the bottom of the table in columns 1, 2, and 4 gives the difference in percentage points in the reduction in lending between loans from the average IRB bank (for which *Share IRB* equals 0.62) and loans from an SA bank (for which *Share IRB* equals zero); the difference between loans from IRB banks and loans from SA banks in column 3; and the difference in percentage points between loans under IRB and loans under SA in columns 7 to 9. Robust standard errors adjusted for clustering at the bank and firm levels are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: <i>Δ Log (loans)</i>								
	Test 1 IRB vs. SA Banks			Test 2 SA Loans of IRB vs. SA Banks			Test 3 IRB vs. SA Loans within IRB Banks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Share IRB</i>	-0.081** (0.018)	-0.053** (0.019)		-0.054** (0.021)	-0.043 (0.035)				
<i>IRB bank</i>			-0.032** (0.014)			-0.030 (0.021)			
<i>IRB loan</i>							-0.039*** (0.011)	-0.040*** (0.014)	-0.021** (0.010)
<i>Portfolio share</i>	0.133 (0.092)	0.111 (0.090)			0.390* (0.211)	0.374** (0.213)		0.145 (0.136)	0.078 (0.095)

Table III—Continued

	Dependent Variable: <i>Δ Log (loans)</i>								
	Test 1 IRB vs. SA Banks			Test 2 SA Loans of IRB vs. SA Banks			Test 3 IRB vs. SA Loans within IRB Banks		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Log bank assets</i>		-0.011** (0.005)	-0.011** (0.005)		-0.007 (0.006)	-0.007 (0.006)		-0.018 (0.013)	
<i>Bank equity ratio</i>		-0.273 (0.336)	-0.171 (0.355)		-0.179 (0.389)	-0.157 (0.391)		0.540 (1.085)	
<i>Bank ROA</i>		-0.008 (0.015)	-0.003 (0.015)		0.016 (0.011)	0.016 (0.011)		-0.107** (0.045)	
<i>State bank</i>		0.007 (0.020)	0.010 (0.020)		0.034* (0.020)	0.037* (0.020)		-0.018 (0.038)	
<i>Cooperative bank</i>		0.009 (0.017)	0.014 (0.016)		0.030 (0.019)	0.033* (0.019)		-0.038* (0.022)	
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	No	No	No	No	No	No	No	Yes
Observations	93,370	93,370	93,370	49,492	49,492	49,492	27,620	27,620	27,620
<i>R</i> ²	0.270	0.270	0.270	0.250	0.251	0.251	0.297	0.299	0.307
IRB impact	-5.0 %	-3.3 %	-3.1 %	-3.3 %	—	—	-3.8 %	-3.9 %	-2.1 %

Economic message. The intensive-margin lending cut is the central result. The extensive-margin evidence is directionally consistent but less robust.

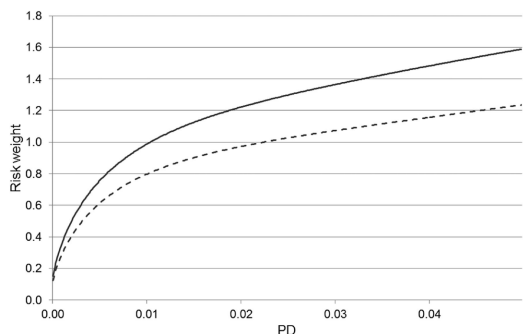


Figure 3. PDs and risk weights. The figure illustrates the relationship between PDs and regulatory risk weights for firms of different sizes. The top line shows the relationship for firms that do not benefit from a discount for SMEs, that is, firms with turnover larger than €50 million. The bottom line shows the relationship for firms with turnover of €10 million.

is more pronounced for loans to which the bank has a large exposure, or for

**Table IV
Extensive Margin**

The table examines the adjustment of lending at the extensive margin. The dependent variable, *Exit*, is a dummy that is equal to one if a loan that existed in the second quarter of 2008 ceased to exist after the credit risk shock. All columns include only loans from IRB banks, and columns 3 and 4 include only firms that have at least one SA loan and at least one IRB loan from an IRB bank. Robust standard errors adjusted for clustering at the bank and firm levels are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: <i>Exit</i>			
	(1)	(2)	(3)	(4)
<i>IRB loan</i>	0.069* (0.039)	0.028 (0.040)	0.044 (0.030)	0.027 (0.024)
<i>Portfolio share</i>	0.569*** (0.214)	0.432 (0.292)	0.163 (0.148)	-0.005 (0.296)
<i>Log bank assets</i>	0.002 (0.013)		-0.042*** (0.011)	
<i>Bank equity ratio</i>	0.796 (1.263)		-0.018 (1.026)	
<i>Bank ROA</i>	-0.016 (0.112)		0.047 (0.068)	
<i>State bank</i>	-0.110*** (0.042)		-0.069** (0.032)	
<i>Cooperative bank</i>	-0.134** (0.057)		-0.097*** (0.034)	
Constant	0.397 (0.364)			
Firm FE	No	No	Yes	Yes
Bank FE	No	Yes	No	Yes
Observations	90,500	90,500	27,620	27,620
<i>R</i> ²	0.038	0.059	0.453	0.465

5.5 Tables V and VI: interest rates and SME discount

Read.

- Table V shows little evidence that IRB loans faced larger interest-rate increases over the shock.
- Table VI shows that IRB loan cuts are larger for firms with higher sales, consistent with the fact that the SME discount dampens the increase in capital charges for smaller firms.
- The size-impact estimates in Table VI are about 1.3 to 3.2 percentage points.

Economic message. Banks respond mainly by reducing loan quantities, not by repricing loans. The SME test strengthens identification because it uses variation in the capital-charge formula within the IRB framework.

5.6 Table VII: heterogeneity by bank capital, loan size, and firm profitability

Read.

- Lower-capital banks cut IRB loans more; the positive interaction with bank equity ratio means the IRB effect is weaker when the bank is better capitalized.
- Large-exposure and large-loan relationships experience much stronger IRB cuts.
- Less profitable firms are hit harder, consistent with larger PD increases for riskier borrowers.

Economic message. The mechanism is strongest exactly where capital constraints and risk-weight increases should matter most.

5.7 Table VIII: firm-level borrowing

Read.

- In the full firm sample, a higher share of IRB loans predicts a larger fall in total firm borrowing.
- In the matched sample with firm controls, the estimated effect is larger.
- The implied reduction is about 3.0 to 11.3 percentage points across specifications, with demand-adjusted effects generally negative.

Economic message. Firms do not fully replace lost IRB credit with other bank credit. The loan-level regulatory channel therefore translates into an aggregate funding effect for firms.

5.8 Table IX: firm capital cost

Read.

- The firm-level cost-of-capital regressions show little robust evidence that firms with more IRB exposure face higher borrowing costs.

Table V
Interest Rates

The table examines how changes in interest rates over the credit risk shock depend on the regulatory approach used for a specific loan. The dependent variable is the difference in *Interest rate* for a specific loan between the pre- and postevent period. Bank controls in column 2 are the same as in Table III. Robust standard errors adjusted for clustering at the bank and firm levels are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: Δ <i>Interest rate</i>			
	(1)	(2)	(3)	(4)
<i>IRB loan</i>	0.0008 (0.0011)	0.0009 (0.0011)	0.0001 (0.0014)	0.0008 (0.0019)
Bank controls	No	Yes	—	—
Bank FE	No	No	Yes	Yes
Firm FE	No	No	No	Yes
Observations	3,132	3,132	3,132	3,132
R^2	0.0003	0.0113	0.0165	0.8795

Table VI
Lending to Small and Medium Enterprises

The table shows the relationship between the reduction in loan size, the regulatory approach used by the bank, and the size of the firm. The dependent variable is the difference in *Log (loans)* between the pre- and postevent periods. The variable *Log(firm sales)* is equal to the logarithm of total sales for firms with turnover less than €50 million and the logarithm of 50 million for firms with a larger turnover; it accounts for the capital charge discount given to loans to SMEs. The size impact at the bottom of the table gives the difference in percentage points in the relative reduction of IRB loans to firms with total sales of €50 million and firms with total sales of €25 million. Robust standard errors adjusted for clustering at the bank and firm levels are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: Δ <i>Log (loans)</i>				
	(1)	(2)	(3)	(4)	(5)
	IRB Loans	SA Loans	All Loans	All Loans	All Loans
<i>Log(firm sales)</i>	-0.009** (0.004)	-0.002 (0.009)	0.006 (0.006)		
<i>IRB loan</i>			0.141 (0.086)	0.340** (0.136)	
<i>IRB loan</i> $\times$ <i>Log(firm sales)</i>			-0.014* (0.008)	-0.035*** (0.013)	-0.026** (0.013)
<i>Constant</i>			-0.104 (0.066)		
Firm FE	No	No	No	Yes	Yes
Bank $\times$ loan pool FE	Yes	Yes	No	No	Yes
Observations	10,374	4,146	14,520	14,520	14,520
R^2	0.011	0.063	0.000	0.517	0.533
Size impact	—	—	-1.3 %	-3.2 %	-2.4 %

Table VII
Firm, Bank, and Loan Characteristics

These regressions examine the interaction between the IRB effect and bank, loan, and firm characteristics. As before, the dependent variable is the difference in *Log (loans)* between the pre- and postevent periods. In columns 1 and 2 we interact *IRB loan* with the *Bank equity ratio* (column 1) or with *High equity*, a dummy that indicates whether the bank equity ratio is above the median (column 2). Bank control variables are the same as in previous tables. In column 3 we interact *IRB loan* with *High exposure*, a dummy that indicates whether a bank's exposure to a specific loan (defined as the loan amount divided by the bank's total assets prior to the shock) is larger than the median, and in column 4 with *Large loan*, a dummy that indicates whether a specific loan is larger than the median loan in our sample. Columns 5 and 6 use a matched sample of 14,336 loans to 3,370 firms for which we are able to obtain balance sheet information and that have at least one SA loan and one IRB loan. We interact *IRB loan* with *Firm ROA* (column 5) or a dummy that indicates whether the firm's pre-shock ROA is higher than the median (column 6). The differential impact at the bottom of the table gives the difference in percentage points in the relative reduction of IRB loans where the respective cross-sectional dummy is equal to one, and IRB loans where it is equal to zero (columns 2 to 4), or the difference between IRB loans, where the respective cross-sectional variable is at its 75th percentile and IRB loans where it is at its 25th percentile (columns 1 and 5). Robust standard errors adjusted for clustering at the bank and firm levels are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: Δ <i>Log (loans)</i>					
	Bank Capitalization		Loan Size		Firm ROA	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>IRB loan</i> $\times$ <i>Bank equity ratio</i>	2.117** (1.039)					
<i>IRB loan</i> $\times$ <i>High equity</i>		0.046** (0.020)				
<i>IRB loan</i> $\times$ <i>High exposure</i>			-0.152** (0.063)			
<i>IRB loan</i> $\times$ <i>Large loan</i>				-0.130*** (0.044)		
<i>IRB loan</i> $\times$ <i>Firm ROA</i>					0.129*** (0.032)	
<i>IRB loan</i> $\times$ <i>High ROA</i>						0.023 (0.018)
<i>IRB loan</i>	-0.102*** (0.030)	-0.064*** (0.018)				
Bank equity measure	0.368 (0.451)	0.040* (0.020)				
Loan size measure			-0.161*** (0.054)	-0.156*** (0.018)		
Bank controls	Yes	Yes	—	—	—	—
Bank $\times$ loan pool FE	No	No	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	Yes
Observations	90,500	90,500	90,500	90,500	14,336	14,336
R^2	0.003	0.004	0.028	0.056	0.363	0.363
Differential impact	3.1 %	4.7 %	-14.1 %	-12.2 %	0.8 %	—

Table VIII
Firm-Level Outcomes

This table reports firm-level results for the full sample of 107,025 firms and for a matched sample of 7,778 firms. As before, we collapse our data into single pre- and postevent time periods. The dependent variable is the change in the logarithm of a firm's total amount of loans around the event. *IRB loan* gives the share of a firm's loans subject to IRB. *IRB loan* gives the share of a firm's loans received from IRB banks subject to IRB. For tests that include this variable the sample is restricted to firms that have at least one IRB loan and at least one SA loan from an IRB bank; that is, firms for which *IRB loan* takes values not equal to zero or one. The IRB impact at the bottom of the table gives the difference in percentage points in the reduction in lending between firms that have only IRB loans and firms that have only SA loans (columns 1 to 4), or between firms where *IRB loan* is at its 75th percentile and firms where *IRB loan* is at its 25th percentile (columns 5 to 8). The bottom of the table also presents coefficients for the IRB variables corrected for firm demand effects by applying the procedure described in Section II.C. Robust standard errors adjusted for clustering at the main bank level are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: Δ <i>Log(firm loans)</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>IRB loan</i>	-0.051*** (0.015)	-0.056*** (0.018)	-0.086*** (0.024)	-0.113*** (0.029)				
<i>IRB loan</i> *					-0.099* (0.057)	-0.117** (0.050)	-0.097** (0.043)	-0.078* (0.040)
<i>Log bank assets</i>	0.006*** (0.001)		0.003** (0.002)		0.003** (0.001)		-0.003 (0.003)	
<i>Bank equity ratio</i>	0.641* (0.335)		-0.334 (0.510)		-1.509* (0.913)		-0.537 (2.001)	
<i>Bank ROA</i>	-0.005 (0.016)		-0.019 (0.024)		-0.004 (0.074)		0.022 (0.088)	
<i>Log firm assets</i>		-0.008 (0.007)	-0.008 (0.007)			0.038*** (0.009)	0.036*** (0.009)	

(Continued)

Table VIII—Continued

	Dependent variable: Δ <i>Log(firm loans)</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Firm ROA</i>			-0.062 (0.059)	-0.070 (0.059)			-0.134 (0.138)	-0.128 (0.138)
<i>Firm leverage ratio</i>			-0.102*** (0.035)	-0.109*** (0.035)			-0.230** (0.115)	-0.188 (0.114)
<i>Constant</i>	-0.060*** (0.004)	-0.214*** (0.041)	0.020 (0.071)	-0.021 (0.083)	-0.118*** (0.031)	-0.125*** (0.044)	-0.545*** (0.101)	-0.475*** (0.100)
Observations	107,025	107,025	7,778	7,778	7,159	7,159	1,575	1,575
R^2	0.003	0.006	0.011	0.012	0.003	0.006	0.026	0.028
IRB impact	-5.1 %	-5.6 %	-8.6 %	-11.3 %	-3.9 %	-4.6 %	-3.8 %	-3.0 %
IRB effect adjusted for demand	-0.046	-0.051	-0.081	-0.108	-0.063	-0.080	-0.062	-0.041

- Only the simplest specification is statistically significant; specifications with bank controls or the restricted IRB-bank share are not.

Economic message. The capital-regulation shock works mainly through credit quantities rather than interest-rate increases.

Table IX
Firm Capital Cost

The table examines how changes in a firm's cost of capital over the event depend on the regulatory approach used for its loans. The dependent variable is the change in a firm's cost of capital (defined as interest paid over total loans) between the pre- and postevent periods. $\overline{IRB\ loan}$ gives the share of a firm's loans that are subject to IRB. The variable $\overline{IRB\ loan}^*$ gives the share of a firm's loans received from IRB banks subject to IRB. For tests that include this variable, the sample is restricted to firms that have at least one IRB loan and at least one SA loan from an IRB bank, that is, firms for which the variable $\overline{IRB\ loan}^*$ takes values not equal to zero or one. Robust standard errors adjusted for clustering at the main bank level are reported in parentheses. * indicates statistical significance at the 10% level, ** at the 5% level, and *** at the 1% level.

	Dependent variable: Δ Firm capital cost			
	(1)	(2)	(3)	(4)
$\overline{IRB\ loan}$	0.0020*** (0.0008)	0.0017 (0.0011)		
$\overline{IRB\ loan}^*$			0.0011 (0.0062)	0.0015 (0.0061)
Firm controls	Yes	Yes	Yes	Yes
Bank controls	No	Yes	No	Yes
Observations	4,977	4,977	1,273	1,273
R^2	0.006	0.010	0.0044	0.0055

6 Short summary

- Basel II IRB regulation makes bank capital requirements depend on internal risk models.
- The Lehman shock raised expected credit risk in Germany.
- IRB capital charges rose, while SA capital charges did not move in the same way.
- German credit-register data let the authors compare IRB and SA loans to the same firms, including within IRB banks.
- IRB loans were cut more than SA loans: the key estimate is a 2.1 to 3.9 percentage point larger reduction.
- The effect is stronger for lower-capital banks, larger exposures, and less profitable firms.
- SME capital-charge discounts reduce the severity of the IRB lending cut, which supports the capital-charge mechanism.
- Firms with more IRB exposure experience larger reductions in total borrowing.
- There is little evidence that banks mainly adjust via interest rates.

7 Conclusion

7.1 Core conclusion

Risk-sensitive capital regulation can amplify downturns. When an economy is hit by a credit-risk shock, model-based capital charges rise for IRB loans. Banks then reduce exactly those loans whose risk weights rise, producing a procyclical lending contraction.

7.2 Economic intuition

The regulation tries to align capital with risk. But in bad times, risk estimates rise precisely when bank capital is scarce. A bank can respond by raising equity, charging higher rates, or cutting assets. The evidence shows that German banks mainly cut IRB loan quantities.

7.3 Takeaway

The paper is a clean empirical demonstration of a macroprudential trade-off: microprudential risk sensitivity improves bank-level measurement of risk, but it can worsen aggregate credit supply when many banks simultaneously deleverage after a common shock.